Homework Set 5

ASTR 5900, Computational Physics & Astrophysics

Due 2026 April 06, 11:59pm

Carry out the following work and write up an informal report showing the outputs (e.g., plots/graphs, tables) and answers to questions. Prepare your report in LaTeX (e.g., using Overleaf), generate a pdf file, and upload your report as a single pdf file on Canvas. Your report should include links to your project code file(s) in your GitHub repository. Code should have adequate comments, be human-readable, and have multiple git commits.

Question 1

1 pt (a) Analytically compute the Fourier Transform $\mathcal{F}$ of the function

$$f(x) = e^{-50(x-\frac{1}{2})^2} \tag{1}$$

Use the convention $\int_{-\infty}^{\infty} f(x)e^{-ikx}dx = \hat{f}(k)$. For reference, the inverse Fourier transform with this convention is $\frac{1}{(2\pi)}\int_{-\infty}^{\infty} \hat{f}(k)e^{ikx}dk = f(x)$
Hint: The following integral may be useful

$$\int_{-\infty}^{\infty} e^{-ax^2+bx}dx = e^{\frac{b^2}{4a}}\sqrt{\frac{\pi}{a}} \tag{2}$$

2 pt (b) Numerically compute the Discrete Fourier Transform of (1) on $[0, 1]$ for three different values of N. Use the conventions

$$X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi\frac{k}{N}n} \tag{3}$$

$$x_n = \frac{1}{N}\sum_{k=0}^{N-1} X_k e^{i2\pi\frac{k}{N}n} \tag{4}$$

for the discrete Fourier transform and the inverse discrete Fourier transform. Plot the three transforms on one plot alongside your analytical result. Invert your k-space results and compare them to (1).

3 pt (c) Use an off-the-shelf FFT algorithm (for example scipy.fft from python) and do the Fourier transform of the function in eq. (1). Compare the FFT and DFT on the same plot. Time the code for your DFT simulation and the FFT simulation with increasing number of grid points. Make a time complexity plot (execution time vs. N) and compare it with the DFT algorithm that you implemented in (b). Use $N = 2^j, j = 3, ..., 11$ for the DFT and $N = 2^j, j = 3, ..., 20$ for the FFT. Since the FFT calculation is very fast, it is susceptible to noise for small N. Average each time over 100 runs for both the DFT and FFT. Plot the times on a log–log plot as a function of N. Comment on your observations.

Question 2

We can use the Fourier transform to solve the 1D heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \tag{5}$$

$$u(x,0) = e^{-50(x-\frac{1}{2})^2}. \tag{6}$$

By using a property of the Fourier transform,

$$\mathcal{F}\left(\frac{d^n f}{dx^n}\right) = (ik)^n \hat{f}(k), \tag{7}$$

the heat equation transforms to

$$\frac{\partial \hat{u}}{\partial t} = -\alpha k^2 \hat{u} \tag{8}$$

$$\hat{u}(k,0) = \mathcal{F}[e^{-50(x-\frac{1}{2})^2}] \tag{9}$$

in k-space.

3 pt

(a) Use the DFT code that you set up in problem 1 to Fourier transform the initial heat distribution (6). Use an Euler/Runge–Kutta numerical integrator to solve (8) in k-space. Use the inverse Fourier transform (4) to go back to position space. Choose $x \in [0,1]$, $\alpha = 0.005$, time step $\Delta t = 0.001$ and total integration time $T = 5$. Plot the solution as a function of space(x-axis) and time(y-axis) with a color plot. Calculate the total heat at each time step, what do you notice?

1 pt

(b) Now choose any other α(say 0.05) and time step (say $\Delta t = 0.01$). What do you observe? Can you comment on the stability of the time-step and α? Comment on your findings.